SPRING END SEMESTER EXAMINATION-2018

8th Semester B.Tech Dual Degree

COMPUTATIONAL INTELLIGENCE

CS-6102

[For 2015(L.E), 2014 & Previous Admitted Batches]

Time: 3 Hours Full Marks: 60

Answer any SIX questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. (a) Draw the schematic diagrams of an expert system and an intelligent system. [2 × 10]

   (b) Define Sugeno's complement and Yager's complement operators. Under what condition, each of these operators can be converted to a classical compoement operator?

   (c) Find the bandwidth of fuzzy set A defined by μA(x) = trapezoid(x; 10, 20, 50, 70).

   (d) Given, μyoung(x)= bell(x;20,2,0) and μold(x) = bell(x; 30,3,100), state the linguistic expression for the term "not young and not very old".

   (e) Explain supervised and unsupervised learning with examples.

   (f) Find the projections of following two-dimensional fuzzy set on X × Y onto X and Y –

$$R = \begin{bmatrix} 0.2 & 0.5 & 0.9 & 0.7 \\ 0.6 & 0.4 & 1.0 & 0.5 \\ 0.8 & 0.1 & 0.3 & 0.6 \end{bmatrix}$$

(g) What is evolutionary computation? What kind of problems is it suitable for?

(h) Given a two-input neuron with parameters: bias(b) = 1.2, weights (W)= [3 2] and inputs (p) = $[-5\ 6]^T$, calculate the neuron output for a Symmetrical Hard Limit transfer function.

(i) Explain the difference between a feed-forward and a recurrent neural network.

(j) What iselitism in GA? Give an example.

2. (a) Explain four types of one-dimensional membership functions with diagrams and associated mathematical expressions. [4]

(b) Define extension principle. Find the image set B for the fuzzy set [4]

$A = 0.1/-3 + 0.2/-2 + 0.8/-1 + 1/0 + 0.4/1 + 0.5/2 + 0.1/3,$

considering the mapping function $y = f(x) = x^2 + 3x - 3$.

3. (a) Mention four basic requirements of T-norm and S-norm operators. What are the four most frequently used T-norm and S-norm operators? [4]

(b) Define composite and non-composite MFs. Find the Cartesian product and Cartesian co-product for the given fuzzy sets – [4]

$A = 0.1/x_1 + 0.4/x_2 + 0.8/x_3 + 1/x_4 + 0.5/x_5$

$B= 0.2/y_1 + 0.3/y_2 + 0.6/y_3 + 0.9/y_4 + 0.7/y_5$

4. (a) Mention eight types of activation functions for neural networks. For a two input single neuron Perceptron, select the weights w_{11} and $w_{12}$and bias b in such a manner that it can function as an AND gate. [4]

(b) Describe the four steps of fuzzy reasoning. Graphically explain the reasoning mechanism for the given cases – [4]

(i) Single rule with multiple antecedent

(ii) multiple rules with multiple antecedents

5. (a) Describe the Sugeno and Tsukamoto Fuzzy models. [4]

(b) Let two relations be defined as, R_1 : "x is near y" on X×Y and R_2 : "y is near z" on Y×Z. R_1 and R_2 are given as [4]

$$R_1 = \begin{vmatrix} 0.2 & 0.5 & 0.9 \\ 0.7 & 0.3 & 0.4 \\ 0.8 & 0.6 & 0.1 \end{vmatrix} \text{ and } R_2 = \begin{vmatrix} 0.8 & 0.3 \\ 0.1 & 0.9 \\ 0.6 & 0.4 \end{vmatrix}$$

Find the relation R_3 : "x is near z" as R_1 o R_2, where o is defined as the max-min composition operator.

6. (a) Prove that the Sugeno's complement operator satisfies the requirements for a truly complement operator. Find the Sugeno's complement for the given fuzzy set [4]

$A = 0.2/x_1 + 0.4/x_2 + 0.7/x_3 + 1/x_4 + 0.6/x_5$

Assume Sugeno parameter, s = 2.

(b) Explain Radial Basis Function Network with suitable diagram. [4]

7. (a) List the characteristics of derivative-free optimization. Briefly explain the components of GA and the steps involved. [4]

(b) Apply the steps of GA to maximize the function [4]

$y = f(x) = 10x\text{-}x^2$ where x is an integer, $0 \leq x \leq 10$.

Show at least 2 steps. Assume suitable values for probability of crossover and probability of mutation.

8. Write short notes on (any two) - [4×2]

(a) ANFIS architecture

(b) Multi-layer perceptron and XOR problem.

(c) Linguistic variable as a quintuple

(d) Three types of universes of discourse

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